

Applicant Tracking System (ATS) - Requirements Document

~Team D

1. Project Overview and Business Context

1.1 Purpose

The ATS addresses critical pain points in the recruitment process:

- Manual collection and handling of applicant data increases operational overhead
- Difficulty in organizing and reviewing multiple applications leads to slower decision-making
- Lack of structured status tracking prevents clear visibility into applicant progress
- Inefficient data extraction for reporting hampers analysis and strategic planning

1.2 System Objectives

1. Automate applicant data collection and management
2. Provide secure, role-based access to applicant information
3. Enable efficient tracking and filtering of applications
4. Facilitate data export for reporting and analysis
5. Ensure data security and controlled access through authentication

1.3 Target Users

- **Applicants:** Candidates applying for Japanese language courses
 - **Administrators:** Staff managing recruitment and application review processes
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2. Functional Requirements

2.1 Applicant Interface Requirements

FR2.1.1: Application Form Page

- Display a single-page or multi-step form for applicants
- Responsive design for mobile and desktop devices
- Clear instructions and field labels for user guidance

FR2.1.2: Personal Information Fields

Applicants must submit the following personal details:

Field Name	Required?	Input Type	Constraints
Full Name	Yes	Text input	2 to 100 characters
Email Address	Yes	Email input	Must be in valid email format
Phone Number	Yes	Numeric input	Must support country code
Current Location	Yes	Text input	No special constraints
Date of Birth	Yes	Date picker	Valid calendar date

FR2.1.3: Educational Background Fields

Applicants must submit the following education details:

Field Name	Required?	Input Type	Constraints
Highest Degree	Yes	Dropdown	High School, Bachelor's, Master's, PhD
Field of Study	Yes	Text input	No special constraints
University / Institution	Yes	Text input	No special constraints
Graduation Year	Yes	Year picker	Valid year
GPA / Percentage	No	Numeric input	0 to 4.0 or 0 to 100

FR2.1.4: Professional Experience Fields

If the candidate has relevant professional experience the following fields must be submitted:

Field Name	Required?	Input Type	Constraints
Years of Experience	Yes	Numeric Input	Range from 0-40
Current Job Title	No	Text input	No special constraints
Company Name	No	Text input	No special constraints
Industry	No	Dropdown	Predefined industry list
Professional Summary	No	Text Area	Maximum 500 characters

FR2.1.5: Course Preference

- Preferred Japanese Language Course (required, dropdown)
- Available options: Beginner, Intermediate, Advanced, Business Japanese
- Course selection must be stored with application

FR2.1.6: CV Upload

- Single file upload field labeled "Upload CV/Resume"
- Accepted formats: PDF, DOC, DOCX
- Maximum file size: 5MB
- File upload is mandatory for form submission
- Display filename and size after upload
- Allow user to replace uploaded file before submission

FR2.1.7: Form Validation

- All required fields must be validated before submission
- Email format validation using RFC 5322 standards
- Phone number format validation
- Display error messages in red text near invalid fields
- Prevent form submission until all validations pass
- Show success/confirmation message after submission

FR2.1.8: Submission Confirmation

- Display confirmation message after successful submission
 - Show application reference/ID for applicant reference
 - Provide expected timeline for application review
 - Option to submit another application or return to home page
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2.2 Admin Interface Requirements

FR2.2.1: Admin Authentication

- Secure login page with email/username and password fields
- Password must be encrypted (minimum 8 characters, mix of uppercase, lowercase, numbers, special characters)
- JWT-based session management with configurable expiration (default: 24 hours)
- "Forgot Password" functionality for account recovery
- Session timeout with warning notification before expiration
- Secure logout functionality

FR2.2.2: Admin Dashboard

- Display all applicants in a responsive table format
- Table columns: Name, Email, Applied Date, Status, Preferred Course, Actions
- Display total count of applications on dashboard
- Display count breakdown by status (New, Under Review, Shortlisted, Rejected)
- Last refresh timestamp visible to admin

FR2.2.3: Application Status Management

- Status field supports four values: **New**, **Under Review**, **Shortlisted**, **Rejected**
- Admin can update status via dropdown in dashboard table
- Status changes must be timestamped and logged
- Visual indicators for status (e.g., color coding: New=Blue, Under Review=Yellow, Shortlisted=Green, Rejected=Red)
- Bulk status update option for multiple applicants

FR2.2.4: Applicant Profile View

- Dedicated profile page showing complete applicant information
- Display all fields: Personal info, education, experience, course preference
- Show submission date and time
- Show current status with timestamp of last status change

- Show action history (status changes with dates)

FR2.2.5: CV Viewing and Download

- In-profile CV preview for supported formats (PDF, DOC, DOCX)
- One-click download button for CV file
- Display file metadata: Upload date, file size, filename
- Secure download link with token-based access (prevent direct file path exposure)

FR2.2.6: Search Functionality

- Search bar on dashboard to query applicants
- Search by Name (partial match, case-insensitive)
- Search by Email (exact match)
- Display matching results in real-time or with search button
- Clear search results option

FR2.2.7: Filtering Functionality

- Multiple filter options accessible via dropdown/checkbox menu:
 - **By Status:** New, Under Review, Shortlisted, Rejected
 - **By Degree:** High School, Bachelor's, Master's, PhD
 - **By Experience Range:** 0-2 years, 2-5 years, 5-10 years, 10+ years
 - **By Preferred Course:** Beginner, Intermediate, Advanced, Business Japanese
- Support for combining multiple filters simultaneously
- Filter count badge showing active filter count
- Clear all filters option
- Apply/Reset buttons for filter control

FR2.2.8: Data Export

- CSV export button on dashboard
- Export all applicant data or filtered results
- CSV columns: Name, Email, Phone, Location, Degree, Field of Study, Experience, Company, Course Preference, Status, Application Date
- File naming: applicants_export_[timestamp].csv
- Export must be instantaneous (< 2 seconds for 1000+ records)

FR2.2.9: Pagination and Data Display

- Table pagination with 10, 25, 50 applicants per page options
 - Navigate between pages with Previous/Next buttons and page indicators
 - Display "Showing X of Y" records
 - Maintain filters and search across pagination
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2.3 Role-Based Access Control

FR2.3.1: Authentication Roles

- **Applicant Role:** Can only view application form and submit applications
- **Admin Role:** Full access to dashboard, profile views, search, filter, export, and status management

FR2.3.2: Access Control Enforcement

- Protect all admin endpoints with JWT token verification
 - Redirect unauthenticated users to login page
 - Redirect applicants attempting admin access to application form
 - Return 401 (Unauthorized) or 403 (Forbidden) HTTP status codes for access violations
 - Token must contain role information for authorization decisions
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3. Non-Functional Requirements

3.1 Performance Requirements

NFR3.1.1: Response Time

- Admin login response: < 2 seconds
- Dashboard load time: < 3 seconds for 1000 applicants
- Search/filter operations: < 1 second response time
- CSV export for 1000 records: < 5 seconds
- File upload response: < 5 seconds for 5MB files

NFR3.1.2: Scalability

- Support minimum 10,000 concurrent applicants
- Database queries must use indexing on Name, Email, Status, Course fields
- API rate limiting: Maximum 100 requests per minute per IP address for admin endpoints

NFR3.1.3: Availability

- System uptime target: 99.5% (excluding scheduled maintenance)
 - Scheduled maintenance window: Monthly, notification 48 hours in advance
 - Graceful error handling with user-friendly error messages
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3.2 Security Requirements

NFR3.2.1: Authentication Security

- Passwords must be hashed using bcrypt (minimum 10 salt rounds)
- JWT tokens must use HS256 or RS256 algorithm
- Token expiration: 24 hours for admin sessions
- HTTPS/TLS 1.2 or higher for all communications
- Secure session management with HttpOnly and Secure cookie flags

NFR3.2.2: Data Protection

- All sensitive data (passwords, tokens) encrypted at rest and in transit
- Input validation on all API endpoints to prevent SQL injection, XSS attacks
- CORS (Cross-Origin Resource Sharing) configured to allow only whitelisted domains
- File uploads validated for file type and size
- Multer configuration: Validate MIME types before processing

NFR3.2.3: Database Security

- IP whitelist configuration can be done using PostgreSQL
- Database connection using connection strings with encrypted credentials
- No sensitive data in logs
- Automated backup every 24 hours with 30-day retention

NFR3.2.4: API Security

- RESTful API endpoints protected with JWT authentication
 - Rate limiting to prevent brute-force attacks
 - API versioning (e.g., /api/v1/)
 - Logging of failed authentication attempts
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3.3 Usability Requirements

NFR3.3.1: User Interface Standards

- Responsive design compatible with mobile (320px), tablet (768px), and desktop (1024px+)
- Consistent styling using Tailwind CSS utility classes
- Standard UI patterns: buttons, forms, tables
- Loading indicators for async operations
- Success/error notification messages using toast or alert system

NFR3.3.2: Accessibility

- WCAG 2.1 Level AA compliance
- Form labels properly associated with input fields
- Keyboard navigation support (Tab, Enter, Escape)
- Alt text for all images
- Color contrast ratio minimum 4.5:1 for normal text

NFR3.3.3: User Experience

- Clear error messages with guidance on resolution
- Confirmation dialogs for destructive actions (delete, bulk status change)
- Form field hints/placeholders for guidance
- Progress indicators for multi-step forms
- Mobile-first design approach

3.4 Maintainability Requirements

NFR3.4.1: Code Quality

- Modular and component-based architecture
- Comprehensive code comments for complex logic
- Consistent naming conventions (camelCase for JavaScript, snake_case for databases)
- DRY (Don't Repeat Yourself) principle adherence
- ESLint configuration for code quality standards

NFR3.4.2: Testing

- Unit test coverage minimum 80%
- Integration tests for critical workflows
- API endpoint testing with Jest/Mocha

- Automated testing in CI/CD pipeline
- Manual user acceptance testing before deployment

NFR3.4.3: Documentation

- API documentation using Swagger/OpenAPI specification
 - Architecture documentation explaining data flow
 - Database schema documentation with ER diagrams
 - Deployment and setup guide for new developers
 - Troubleshooting guide for common issues
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4. Technical Requirements

4.1 Frontend Stack

- **Framework:** Next.js 14+ (React-based)
- **Styling:** Tailwind CSS for utility-first CSS
- **Form Handling:** React Hook Form for form validation and submission
- **HTTP Client:** Axios for API communication
- **State Management:** React Context API or Redux Toolkit (if complex state needed)
- **Build Tool:** Next.js built-in Webpack configuration

4.2 Backend Stack

- **Runtime:** Node.js 18+
- **Framework:** Express.js
- **API Style:** RESTful API with JSON request/response
- **Authentication:** JSON Web Tokens (JWT)
- **Middleware:** Express middleware for logging, error handling, CORS

4.3 Database

- **Primary Database:** PostgreSQL
- **Indexing:** Indexes on Name, Email, Status, Course fields for query optimization

4.4 File Storage

- **Service:** Amazon AWS S3
- **Configuration:** Separate bucket for development, staging, production
- **Access Control:** IAM roles with least privilege
- **File Management:** Multer for handling multipart form uploads
- **Path Structure:** applicants/{applicationId}/cv.{extension}

4.5 Authentication and Authorization

- **Method:** JWT with role-based access control (RBAC)
- **Token Management:** Refresh token rotation (optional for 24-hour expiration)
- **Libraries:** jsonwebtoken for JWT creation/verification

4.6 Supporting Libraries

- **CSV Export:** json2csv for applicant data export
- **File Handling:** Multer for file uploads
- **Configuration:** dotenv for environment variables
- **Logging:** Morgan or Winston for request/error logging
- **Validation:** Joi or express-validator for input validation